

TABLE 38 ASIA PACIFIC: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Japan	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
India	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Rest of Asia Pacific	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Total	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 39 LATIN AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Rest of Latin America	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Total	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.4 SHOTGUN SEQUENCING

7.4.1 ADVANCEMENTS IN DATA ANALYSIS SOLUTIONS TO SUPPORT ADOPTION

Shotgun sequencing involves breaking up large DNA sequences into multiple small pieces/sequences and then reassembling them into larger contiguous pieces representing the original large DNA sequence. Shotgun sequencing is a relatively new approach used to analyze many organisms present in a given complex sample and comprehensively sequence all genes. This technique provides researchers or end users with the necessary information to evaluate community biodiversity and its related function in the ecosystem.

The advantages offered by shotgun metagenomic sequencing over capillary sequencing, PCR-based approaches, and traditional next-generation sequencing techniques, the growing adoption of shotgun metagenomic sequencing among end users, and the growing number of metagenomic sequencing research activities across the globe are the key factors driving the growth of the shotgun metagenomic sequencing market.

Some of the advantages of shotgun metagenomic sequencing are mentioned below:

- Provides tools to study unculturable microorganisms that are difficult or impossible to analyze
- Eliminates the mapping stages, resulting in faster processing and whole-genome sequencing than clone-by-clone sequencing
- Ability to combine many samples in a single sequencing run and obtain high sequence coverage per sample
- Can detect very low abundance organisms of a microbial community, which may be missed or are too expensive to identify using other traditional methods

The challenges related to data interpretation and analysis in shotgun metagenomic sequencing and the high cost associated with the sequencing of multiple genomes are the key factors restraining the growth of this market. On the other hand, the rapid advancements and developments in informatics software are expected to improve the ease and efficiency of shotgun metagenomic analysis.

TABLE 40 SHOTGUN SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%
Europe	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%
Asia Pacific	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%
Latin America	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%
Middle East and Africa	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Total	59.09	67.70	77.52	88.83	101.89	117.27	135.44	157.10	15.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 41 NORTH AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	30.64	34.90	39.74	45.29	51.65	59.11	67.88	78.29	14.5%
Canada	3.17	3.58	4.05	4.57	5.18	5.88	6.70	7.67	13.6%
Total	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 42 EUROPE: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.96	4.57	5.26	6.06	6.99	8.09	9.40	10.96	15.8%
UK	3.54	4.11	4.75	5.51	6.39	7.43	8.67	10.17	16.4%
France	2.80	3.22	3.71	4.27	4.92	5.69	6.60	7.69	15.7%
Italy	0.79	0.91	1.04	1.19	1.37	1.59	1.83	2.13	15.4%
Spain	0.61	0.70	0.80	0.91	1.05	1.21	1.40	1.62	15.3%
Rest of Europe	6.60	7.57	8.67	9.95	11.42	13.14	15.18	17.62	15.2%
Total	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 43 ASIA PACIFIC: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.13	2.52	2.99	3.53	4.18	4.95	5.89	7.02	18.7%
Japan	0.99	1.18	1.41	1.67	1.99	2.37	2.84	3.41	19.3%
India	0.13	0.15	0.18	0.21	0.25	0.29	0.35	0.41	18.5%
Rest of Asia Pacific	1.79	2.12	2.50	2.95	3.48	4.12	4.89	5.82	18.5%
Total	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 44 LATIN AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.54	0.61	0.68	0.77	0.86	0.97	1.09	1.24	12.7%
Rest of Latin America	0.61	0.68	0.77	0.86	0.97	1.09	1.23	1.39	12.6%
Total	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.5 OTHER TYPES

This segment is inclusive of de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing. Metatranscriptomics is a key technique in metagenomics, which allows researchers to study the function and activity of a complete set of transcripts (RNA sequences) from environmental samples.

The growing number of metagenomic research activities is the major factor driving the growth of this market segment. Over the last few years, the focus on understanding the function and relation of gut microbiome diversity, predicting resistance to specific antibiotics, understanding host-pathogen immune interactions, quantifying gene expression changes, and tracking disease progression has increased significantly.

De novo sequencing generates an initial genomic sequence of a particular organism without a reference genome. De novo sequencing services are applied in the research of animals, plants, and microorganisms, including phylogenetic studies, analysis of species diversity, genetic markers, and other genomic research. Using de novo sequencing to obtain the genomic information of microbes provides a fresh start for exploring the genetic structure and functions, studying the evolutionary origin of microbial populations, as well as developing potential applications across medicine, disease modeling, agriculture, and the environment.

TABLE 45 OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%
Europe	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%
Asia Pacific	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Latin America	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Middle East and Africa	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 46 NORTH AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	11.23	12.59	14.11	15.82	17.76	20.00	22.60	25.65	12.7%
Canada	1.16	1.29	1.43	1.59	1.77	1.97	2.20	2.47	11.5%
Total	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis